# L2.8 Antiderivatives

§3.9 — run the rules backwards, then pin down the constant

Every answer with a $+C$ in it can be checked the same way: differentiate what you wrote and see whether you get back what you started with.

## 1 · The reverse power rule

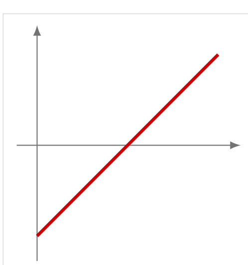
**1.** Find the antiderivative of each. **Rewrite every term as a power first.**

$$\text{(a)}\ \ 3\sqrt{x} - 2\sqrt[3]{x} \qquad \text{(b)}\ \ (x-5)^2 \qquad \text{(c)}\ \ \frac{10}{x^9}$$

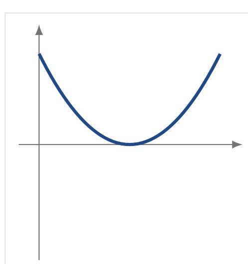
**Stop.** There is no reverse chain rule and no reverse product rule. Part (b) has to be expanded before any rule applies.

## 2 · Reading an antiderivative off a graph

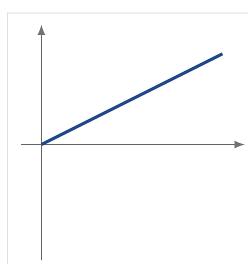
**2.** The graph of $f$ is on the left. **Which of the three graphs beside it is an antiderivative of $f$, and how do you know?** Say what rules out each of the other two.



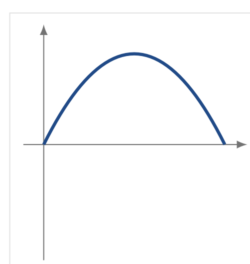
**the graph of $f$**



**(a)**



**(b)**



**(c)**

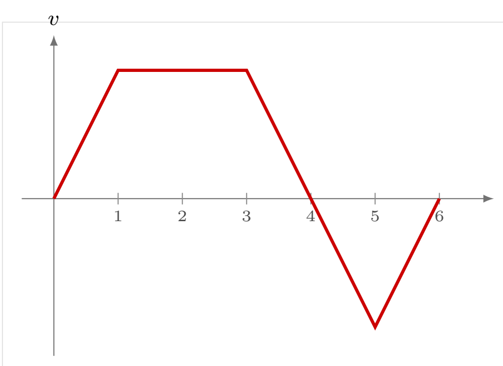
## 3 · Initial value problems

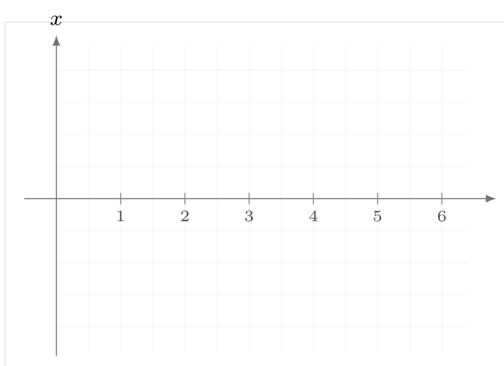
**3.** $f''(x) = 12x^2 - 6x + 2$, with $f'(1) = 5$ and $f(1) = -1$. Find $f$.

**Stop.** Two antidifferentiations, two constants. Resolve the first one *before* you antidifferentiate again.

## 4 · Position from acceleration

**4.** The velocity of an object is graphed below, and the object starts at $x(0) = 0$. **Sketch its position** on the empty axes.




Mark the $t$ where $x$ is largest, and say how you knew.

**5.** An object moves with **constant** acceleration $a$. Its velocity at time 0 is $v_0$ and its position at time 0 is $x_0$. Show that

$$x = \frac{1}{2}at^2 + v_0 t + x_0$$